Renke Wang

Fairfax, VA — renke.wang.ece@gmail.com — Website — LinkedIn — Google Scholar

Professional Summary

Ph.D. candidate in Electrical and Computer Engineering working on scheduling–control co-design and decision-making for multi-agent robotic systems that compete for limited shared resources (physical space, computing, and operator attention). Combines Model Predictive Control (MPC) with decision-tree search for exact scheduling, and develops distributed optimization and deep reinforcement learning frameworks for large-scale systems. Also works on Lyapunov-based stability analysis and controller design, including event-triggered control, for positive, switched, and networked control systems under additive disturbances and timing uncertainty.

Education

George Mason University *Expected 2027*
Ph.D. Candidate in Electrical and Computer Engineering
Dalian University of Technology
Master of Engineering, Control Science and Engineering
Dalian University of Technology
Bachelor of Engineering, Automation

Technical Skills

Planning: multi-robot scheduling, path planning, decision-tree search, graph search (Dijkstra, A*).

Control: MPC, robust MPC, Lyapunov- and LMI-based stability analysis, event-triggered control.

Optimization: linear and mixed-integer programming (LP, MILP), distributed optimization (consensus ADMM).

Learning: deep reinforcement learning (PPO), supervised learning.

Programming & Tools: Python, MATLAB, C++, PyTorch, CasADi, Gurobi, YALMIP, Git, Linux, LaTeX.

Research Experience

Research Assistant at George Mason University *Fairfax, VA*
August 2022 – Present

Multi-Agent Coordination under Shared Resources

- **Shared Physical Space:** Formulated contention-resolving MPC (CMPC), combining decision-tree search over access order with finite-horizon MPC in CasADi, for vehicles and robots at multi-lane intersections, warehouse merging zones, and multi-intersection networks; evaluated trajectories, resource utilization, and computation time in event-driven simulation (CCTA 2025).
- **Shared Computing Resources:** Co-designed processor allocation, task scheduling, and motion control for multiple blimps sharing heterogeneous processors for remote wind estimation; modeled processor-dependent execution times, scheduled processor access via decision-tree search, and enforced delay bounds under wind disturbances to keep trajectories feasible (IJRNC, under review).
- **Shared Operator Attention:** Developed the real-time scheduler of a recommendation system that tells a human operator which robot to assist and when during time-critical obstacle avoidance, modeling interventions as deadline-constrained tasks solved by decision-tree search; the system was evaluated in a 40-participant user study (CCTA 2024, RO-MAN 2025).

Learning-Based Scheduling and Path Selection

- Trained a supervised neural scheduling policy that imitates exact CMPC tree-search decisions.
- Developing event-driven PPO actor-critic frameworks for shared-resource scheduling and for joint path selection and scheduling, extending fixed-route coordination to online routing.

Distributed Scheduling and Control Co-Design for Networked Control Systems

- Developed a distributed framework for control tasks sharing sensing, computation, and actuation resources: consensus ADMM coordinates task timing across resource-level subproblems, each solved in parallel by local CMPC (TCNS, under review).

– Benchmarked closed-loop cost, computation time, and scalability against EDF and centralized CMPC, with a 50-restart sensitivity study and a CACC platoon case study.

Robust Control under Distribution-Free Timing Perturbations

– Developed a tube-based robust MPC for LTI systems under distribution-free control-update timing perturbations; designed timing-dependent dynamic feedback gains, derived the closed-loop error dynamics, and proved asymptotic stability and the existence of a forward invariant set (ACC 2024).

Graduate Researcher at Dalian University of Technology *Dalian, China*
Sept 2017 – June 2020

Stability Analysis and Controller Design for Positive and Switched Systems

– Derived Lyapunov-based stability conditions for positive switched systems and Markov jump linear systems (MJLS) under dwell-time constraints and fully or partially known transition rates.

– Designed state-feedback and L1-gain controllers using piecewise-linear copositive Lyapunov functions, establishing stochastic and exponential stability under external disturbances and controller failures (NAHS 2021).

Selected Publications and Manuscripts

- Renke Wang and Ningshi Yao. *Distributed Scheduling and Control Co-Design for Networked Control Systems with Multi-Layered Shared Resources.* IEEE Transactions on Control of Network Systems (TCNS), under review.
- Renke Wang, Mengxue Hou, and Ningshi Yao. *Robust Co-Design of Scheduling and Control for Lighter-than-Air Agents with Heterogeneous Resources.* International Journal of Robust and Nonlinear Control, under review.
- J. Lian and R. Wang. *Stochastic Stability of Positive Markov Jump Linear Systems with Fixed Dwell Time.* Nonlinear Analysis: Hybrid Systems, 40 (2021): 101014.
- R. Wang and N. Yao. *Contention-Resolving Model Predictive Control for Coordinating Automated Vehicles at a Multi-Lane Traffic Intersection.* 2025 IEEE Conference on Control Technology and Applications (CCTA). IEEE, 2025.
- Rajul Kumar, Renke Wang, and Ningshi Yao. *One Human and Multi-Robot Collaboration: Evaluating the Impact of Attention-Guided On-Screen Recommendations on Human Multitasking Efficiency, Cognitive Load, and Trust.* 2025 34th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), IEEE, 2025.
- R. Wang and Ningshi Yao. *Robust Model Predictive Control for Networked Control Systems with Timing Perturbations.* 2024 American Control Conference (ACC). IEEE, 2024.
- N. Yao and R. Wang. *Optimal Real-Time Human Attention Allocation and Scheduling in a Multi-human and Multi-robot Collaborative System.* 2024 IEEE Conference on Control Technology and Applications (CCTA). IEEE, 2024.

Teaching Experience

Teaching Assistant, George Mason University *Fairfax, VA*
Aug 2022 – Dec 2022

- Delivered lectures and led or co-led laboratory sessions for Continuous-Time Signals and Systems; explained technical concepts, guided laboratory work, graded assignments, and supported undergraduate students.
- Graded coursework and answered student questions for Classical Systems and Control Theory, providing clarification on course concepts and assignments.

Honors and Professional Service

- Summer Research Assistantship Award, George Mason University Graduate Division, 2026.
- Reviewer for IEEE Transactions on Control Systems Technology; IEEE Transactions on Systems, Man, and Cybernetics: Systems; and IEEE conferences including ACC, CDC, and CCTA, 2023–Present.